

CSA5103 - CRYPTOGRAPHY AND NETWORK SECURITY LIST OF PROGRAMS

EXPERIMENT 1

Write a C program for Caesar cipher involves replacing each letter of the alphabet with the letter standing k places further down the alphabet, for k in the range 1 through 25.

Aim: To write a Python program to implement the Caesar Cipher technique for encrypting a given plaintext using a user-defined key (1–25).

Algorithm:

Step 1: Start.

Step 2: Read the plaintext and the key from the user.

Step 3: Initialize an empty string encrypted.

Step 4: Process each character in the plaintext.

Step 5: If the character is an alphabet, encrypt it using the key; otherwise, keep it unchanged.

Step 6: Store the encrypted characters in encrypted.

Step 7: Display the encrypted text.

Program:

```
text = input("Enter the text: ")
key = int(input("Enter the key (1-25): "))
encrypted = ""
for ch in text:
    if ch.isupper():
        encrypted += chr((ord(ch) - ord('A') + key) % 26 + ord('A'))
    elif ch.islower():
        encrypted += chr((ord(ch) - ord('a') + key) % 26 + ord('a'))
    else:
        encrypted += ch
print("Encrypted Text:", encrypted)
```

Output:

```
=====
Enter the text: HELLO
Enter the key (1-25): 3
Encrypted Text: KHOOR
=====
Enter the text: YELLOW
Enter the key (1-25): 1
Encrypted Text: ZFMMPX
=====
```

Result:

The Caesar Cipher encryption program was successfully implemented, and the plaintext was converted into ciphertext using the given encryption key.